

The Verification Gap: Why AI-Augmented Academic Writing Needs Reporting Guidelines for Reasoning

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Abstract

Large language models are increasingly used to assist with academic writing. While citation hallucination has received considerable attention, AI writing assistants create additional verification challenges—confidence inflation, scope creep, and fluent prose that obscures reasoning gaps—that existing tools do not address. This gap is compounded by a structural absence in reporting guidelines: the EQUATOR Network maintains nearly 700 guidelines for empirical health research, yet no equivalent exists for non-empirical paper types. Drawing on Gregor’s (2006) theory-type taxonomy, we identify two fundamentally non-empirical theory types for which argument quality, rather than factual accuracy, is the primary verification challenge. We propose a typed verification model distinguishing claims (source-verifiable facts), arguments (Toulmin-verified reasoning chains), and propositions (Whetten-verified recommendations with boundary conditions), each requiring distinct procedures. We propose that the scholarly community develop verification infrastructure for non-empirical papers analogous to what EQUATOR provides for empirical research.

Keywords: academic writing, artificial intelligence, reporting guidelines, verification, argumentation, non-empirical research

Key Points:

- AI writing tools create verification challenges beyond citation hallucination, including confidence inflation and argument degradation.
- EQUATOR provides nearly 700 guidelines for empirical research; non-empirical papers have no equivalent infrastructure.
- We propose typed verification distinguishing claims, arguments, and propositions, each with distinct procedures.
- Wrong-type verification can produce false failures invisible to conventional approaches.

1 Introduction: The Problem

The rapid adoption of large language models (LLMs) as writing assistants is transforming how academic papers are produced. Liang et al. (6) found that up to 22% of computer science papers now show evidence of LLM modification, with detectable usage across all major disciplines. These tools can synthesise literature, structure arguments, generate prose, and format references with remarkable fluency. Yet this fluency introduces risks that are distinct from—and in some ways more subtle than—the challenges of traditional academic writing. Where a human author’s limitations are typically visible in the text (awkward phrasing, gaps in coverage, inconsistent formatting), an AI-assisted draft can be polished and comprehensive while harbouring verification failures that are invisible to casual inspection.

The most widely discussed risk is citation hallucination. LLMs can invent plausible-sounding papers, complete with fabricated authors, journal names, and digital object identifiers. Mugaanyi et al. (7) found that 62–89% of DOIs generated by large language models across natural sciences and humanities topics were fabricated, with DOI accuracy as low as 8.5% in some disciplines. Unlike factual errors that contradict known evidence, hallucinated citations create a distinctive category of falsehood: references that cannot be falsified through domain knowledge alone because they refer to works that do not exist. Citation hallucination has received considerable attention, and tools designed to detect it are emerging.

However, citation hallucination may be only the most visible failure mode. A second, potentially more insidious risk is what we term *confidence inflation*: AI writing assistants may state speculative claims with the same certainty as well-established findings, using language such as “demonstrates” or “confirms” where “suggests” or “may indicate” would be more appropriate. In academic writing, the distinction between “demonstrates” and “suggests” carries substantial epistemic weight—it signals to readers whether a claim rests on strong evidence or preliminary findings. When AI-generated prose conflates these registers, the resulting text can misrepresent the state of evidence without containing any factually incorrect statements. Liang et al. (5) found that LLM-generated feedback

on research papers catches surface issues but struggles with the kind of deep argument analysis that would detect such confidence miscalibration.

A third risk applies specifically to technical and empirical papers: AI-generated equations and derived values can contain arithmetic errors that survive review because they produce plausible-looking results. Errors that produce numbers in physically reasonable ranges can survive plausibility review even when the underlying arithmetic is incorrect. This suggests that for quantitative content, *mechanical reproduction* (substituting values and computing) catches errors that *expert assessment* (evaluating whether the physics looks right) does not.

A fourth risk is scope creep. Without structural constraints such as section specifications and word budgets, AI-assisted drafts may expand arguments beyond what the available evidence supports, introduce tangential sections, and exceed format requirements. While scope management is a challenge in all academic writing, AI tools can amplify it by generating additional content with minimal effort, making expansion the path of least resistance.

These failure modes share a common characteristic: they cannot be detected by checking whether citations are real. Citation verification tools that automatically check whether references exist, along with model-level approaches such as retrieval-augmented generation (in which the model is given access to source documents during text generation), address the factual accuracy dimension of AI-assisted writing. These are valuable contributions, but they operate at the model or tool level—they improve what the AI generates or verify whether specific references exist. To our knowledge, no process-level verification infrastructure addresses the writing workflow itself: the systematic registration, tracking, and confidence calibration of claims throughout the drafting process.

The distinction between tool-level and process-level verification matters because the failure modes described above are not primarily failures of the AI model. They are failures of the writing process—failures that occur when there is no infrastructure to register claims at appropriate confidence levels, calibrate language to evidence strength, or verify that arguments are logically sound rather than merely well-sourced. This paper

asks what such process-level infrastructure might look like and whether existing scholarly frameworks provide a foundation for building it.

2 The Landscape Gap

To assess the current state of verification infrastructure for academic writing, it is useful to examine the most comprehensive effort to date: the EQUATOR Network. EQUATOR (Enhancing the QUALity and Transparency Of health Research) maintains a database of nearly 700 reporting guidelines for health research (EQUATOR Network). These guidelines—including CONSORT for randomised controlled trials, STROBE for observational studies, and PRISMA for systematic reviews (8)—provide structured checklists that authors complete during manuscript preparation. The GoodReports tool (10) automates guideline selection based on study type, further reducing the barrier to adoption. A Cochrane review found that journals endorsing CONSORT showed improved reporting completeness across the majority of checklist items (13), and incomplete reporting has been identified as a major contributor to research waste (2). These guidelines represent the most mature infrastructure for structured verification in academic publishing.

However, an examination of the EQUATOR database reveals a structural gap. Virtually all of its nearly 700 guidelines address empirical research: studies that collect, analyse, and report data. Systematic reviews, covered by PRISMA (8), represent a partial exception—they are structured and methodical but synthesise rather than generate primary data. Beyond this partial exception, to our knowledge, no EQUATOR or equivalent guidelines exist for theoretical papers, design science contributions, perspective articles, or methodological papers that do not report empirical findings.

This absence is not accidental. It reflects a fundamental difference in what needs to be verified across paper types. Gregor (3) identified five theory types in information systems research based on four goals: analysis, explanation, prediction, and prescription. Two of these types—Type I (analytic theories, including taxonomies and frameworks) and Type V (design and action theories, including design principles and methods)—are fundamentally non-empirical in their primary contribution, though Type V papers may

include empirical evaluation of the designed artefact. Their core contributions are not data points that can be verified against measurements or observations; they are conceptual structures whose value lies in their clarity, coherence, and utility. The remaining types (Type II: explanatory, Type III: predictive, Type IV: explanatory and predictive) align more naturally with empirical verification approaches, as they make claims about observable phenomena.

While Gregor’s taxonomy was developed for information systems, the underlying distinction applies broadly. The IS community has developed structured methodologies for non-empirical work—notably Peffers et al.’s (2007) widely adopted design science research process model—yet these methodological frameworks have not been translated into reporting guidelines analogous to EQUATOR. More broadly, perspective articles, commentaries, theoretical syntheses, and methodological proposals all share the characteristic that their primary contribution is an argument or a framework rather than empirical evidence. These paper types appear across virtually every academic discipline, from the humanities and social sciences to engineering and medicine.

For these non-empirical paper types, the central verification challenge may not be “do the data support the claims?” but rather “is the argument logically sound, are the premises established, and are the boundary conditions specified?” (12; 14). This is a fundamentally different verification task from the one that EQUATOR guidelines are designed to support.

This suggests that the EQUATOR gap is structural rather than a mere oversight. Empirical papers need verification of research design, data collection, analysis, and reporting—domains for which structured checklists are well-suited. Non-empirical papers need verification of argument quality: warrant validity, reasoning coherence, confidence calibration, and boundary specification. These are different verification tasks that may require different infrastructure. While this gap is most acute in purely non-empirical papers, it represents a systemic risk wherever authors move from data-driven claims to interpretive reasoning—including the discussion sections of empirical research. The question is whether established frameworks from argumentation theory and theory-building method-

ology can provide a foundation for this infrastructure.

3 Typed Verification: A Proposal

We propose that verification infrastructure for non-empirical papers should distinguish three unit types, each requiring a distinct verification procedure (Table 1).

Table 1: Three verification unit types with per-type procedures.

Type	Definition	Verification question	Framework
Claim	Source-verifiable fact	Does the source say this?	Citation checking
Argument	Evidence + inference $\rightarrow$ conclusion	Is the reasoning valid?	Toulmin (1958/2003)
Proposition	Recommendation with boundaries	Are conditions specified?	Whetten (1989)

The first and most familiar type is the **claim**: a factual statement that can be verified against one or more sources. Claims are the default unit in citation-based verification—“Study X found Y” is a claim whose accuracy can be checked by reading Study X. Most existing verification tools, including citation checkers and fact-verification systems, operate on claims. For claims, the verification question is straightforward: does the cited source exist, and does it say what the manuscript attributes to it?

The second type is the **argument**: a reasoning chain that combines evidence with inference to reach a conclusion. Arguments appear throughout academic writing, particularly in discussion and conclusion sections, yet they resist claim-level verification because their validity depends not on any single source but on the logical relationship between premises and conclusion. Toulmin (12) provides a well-established framework for analysing argument structure, decomposing any argument into six components: claim (the conclusion asserted), grounds (the evidence offered), warrant (the inferential bridge connecting grounds to claim), backing (support for the warrant), qualifier (the degree of certainty), and rebuttal (conditions under which the argument does not hold). This framework may provide a basis for operationalising argument verification in academic writing: an argument can be evaluated by checking whether its grounds are established, its warrant is explicit and valid for the target audience, and its qualifier is calibrated to the strength of the evidence. (Our operationalisation focuses on five of these six com-

ponents, treating backing as implicit when the warrant itself is accepted by the target audience.) Recent work by Gupta et al. (4) found that LLMs can extract Toulmin-style argument structures through zero-shot prompting, suggesting that automated support for argument-level verification may be technically feasible.

The third type is the **proposition**: a novel recommendation or contribution that specifies what should be done and under what conditions. Propositions are the defining unit of many non-empirical papers—they are what the paper contributes to the field. Whetten (14) provides a framework for evaluating theoretical contributions through four components: What (the constructs or factors involved), How (the relationships between them), Why (the underlying logic or reasoning), and Who/Where/When (the boundary conditions specifying where the contribution applies). Propositions can be evaluated by checking whether all key constructs are defined, whether the reasoning is explicit, and whether boundary conditions are stated—without requiring empirical data.

These three types may benefit from a shared confidence calibration system. We propose mapping confidence tiers to prescribed language (Table 2), so that the certainty expressed in the text is systematically aligned with the strength of the underlying evidence. This mapping may enable authors and reviewers to detect confidence inflation—the failure mode described in Section 1—at the drafting stage rather than during peer review.

Table 2: Proposed confidence-to-language mapping.

Tier	Assign when...	Prescribed language
Established	Multiple independent sources; no dispute	“demonstrates,” “confirms”
Supported	2–3 sources agree; open questions remain	“indicates,” “evidence suggests”
Emerging	1–2 sources; not yet replicated	“may,” “preliminary evidence”
Speculative	Logical inference; no direct data	“warrants investigation”

The central argument of this paper is that these three types require fundamentally different verification procedures, and that applying the wrong procedure to the wrong type may produce systematic errors. A claim evaluated as an argument (checking its warrant rather than its source) wastes effort. More consequentially, an argument evaluated as a claim (checking whether it has a source rather than whether its reasoning is valid) may produce false failures—entries scored as unverified not because they are wrong, but

because the verification procedure was inappropriate for the unit type. This argument rests on the premise that different epistemic contributions (facts, inferences, recommendations) have different validity criteria; if this premise holds, then a unified verification procedure will systematically misclassify some entries. One might question whether a three-type model is necessary—whether a simpler binary distinction between empirical and non-empirical verification would suffice. However, the distinction between arguments and propositions matters because they fail differently: an argument with a weak warrant needs better reasoning, while a proposition with missing boundary conditions needs better scoping. Conflating the two under a single “non-empirical” category obscures which corrective action is appropriate.

4 Related Work and Design Rationale

The framework described in Section 3 is proposed as a process-level addition to existing tool-level and model-level approaches; an empirical evaluation across multiple paper projects, authors, and domains is needed before generalisable claims about its effectiveness can be made. A self-application data point is provided in Appendix A—the verification registry maintained during the writing of this paper, showing the framework applied prospectively to a perspective-paper contribution.

The design rationale draws on independent research on structured verification for AI-assisted scholarly work. Sukpanichnant et al. (11) developed PeerArg, a framework that combines formal argumentation structure with LLM-based review analysis; their results suggest that structured representation of support and attack relations improves review aggregation compared to end-to-end LLM approaches. Similarly, Gupta et al. (4) found that prompting LLMs with Toulmin’s argument model significantly improves argument extraction compared to generic prompts. Both findings suggest that combining structured frameworks with LLM capabilities may outperform LLM capabilities alone—a principle consistent with the process-level verification approach proposed here.

The proposal is offered as a model for the scholarly community to test, adapt, and evaluate. Whether typed verification provides genuine value across authors, domains, and

writing workflows remains an open empirical question.

5 Call to Action

AI writing assistance creates verification challenges that extend beyond citation hallucination (Section 1), yet EQUATOR’s nearly 700 reporting guidelines address empirical research exclusively—leaving non-empirical paper types without systematic verification support (Section 2). On the basis of this gap, and preliminary evidence that typed verification reveals actionable issues invisible to conventional approaches (Section 4), we propose that the scholarly community develop verification infrastructure for non-empirical papers. Specifically, we propose that verification procedures be differentiated by unit type:

- **Source verification** for claims (does the cited source say this?);
- **Toulmin-based analysis** for arguments (is the reasoning valid, are the premises established, is the qualifier calibrated?);
- **Whetten-based analysis** for propositions (are the boundary conditions specified, is the underlying logic explicit?).

Each type would have associated checklists, confidence calibration criteria, and language guidelines that authors can apply during manuscript preparation.

This proposal has specific boundary conditions. It applies to academic papers intended for peer-reviewed publication in fields where argument quality constitutes a primary contribution. It does not apply to informal writing, journalism, creative writing, or very short opinion pieces where the overhead of formal verification would exceed the benefit. Furthermore, the specific verification frameworks proposed here—Toulmin’s argument model and Whetten’s theoretical contribution framework—are grounded in Western academic argumentation traditions. Their applicability to scholarly traditions with different rhetorical conventions would need to be assessed rather than assumed.

One might object that journal peer review already provides adequate argument verification for non-empirical papers—that reviewers routinely evaluate logical coherence, boundary conditions, and counter-argument engagement. This observation has merit, and peer review remains essential. However, AI writing assistance may amplify the verification challenge beyond what traditional peer review was designed to handle. When AI tools can generate fluent, well-structured prose with minimal effort, the volume and apparent quality of submitted manuscripts may increase while the ratio of well-reasoned to superficially coherent arguments decreases. Reviewers already face growing workloads; adding the burden of detecting AI-generated confidence inflation and implicit warrant failures may exceed what the current system can absorb. Process-level verification infrastructure would function as a complement to peer review, not a replacement—catching argument-level issues before submission so that reviewers can focus on substantive evaluation rather than structural verification.

Several directions warrant investigation. First, the typed verification model proposed here has been developed and applied within medical simulation and instrumentation. Discipline-specific verification templates—adapting the three-type model to the conventions of different fields—would be needed to assess the framework’s generalisability. Second, integration with AI writing tools could enable real-time verification during the drafting process, using the structured checklists and confidence calibration criteria as prompting infrastructure for writing assistants. Third, empirical validation comparing the quality of papers written with and without typed verification infrastructure would provide the evidence base needed to move the proposal from preliminary to supported. Fourth, while this paper focuses on non-empirical paper types, the discussion sections of empirical papers involve the same argumentative move from data to interpretation—and may benefit from the same typed verification procedures applied to that transition. Fifth, the calculation errors described in Section 1—where AI-generated equations produce plausible but incorrect results—suggest that numerical reproduction (mechanically recomputing derived values from stated inputs) may warrant recognition as a distinct verification procedure alongside source checking, argument analysis, and proposition evaluation.

The EQUATOR Network has spent two decades building nearly 700 reporting guidelines for empirical health research. These guidelines have measurably improved the quality and transparency of published research (13; 2). To our knowledge, essentially no equivalent guidelines exist for non-empirical papers—theoretical contributions, design science research, perspective articles, and methodological proposals. As AI writing tools make it easier to produce fluent academic prose, the gap between what is published and what is rigorously verified may widen. We suggest it is time to begin building the verification infrastructure for the rest.

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[To be added.]

A Self-Application of the Typed Verification Framework

This paper uses the framework it describes. The claim registry for this manuscript was maintained during writing—entries were registered, typed, and verified as the paper was drafted. This appendix summarises that registry to provide a prospective data point that reviewers can independently verify.

Registry summary

Table 3 presents all 19 registered entries. The distribution across types—16 claims, 2 arguments, 1 proposition—reflects the paper’s structure: a perspective article grounded primarily in verifiable facts and reasoning chains, culminating in a single recommendation.

Worked examples by type

Claim (S2-1). “The EQUATOR Network maintains nearly 700 reporting guidelines.” This entry was initially drafted as “approximately 500 guidelines” based on a secondary

Table 3: Complete verification registry for this manuscript (19 entries).

ID	Type	Priority	Confidence	Status
S1-1	Claim	P0	Supported	Verified
S1-2	Claim	P0	Supported	Verified
S1-3	Claim	P1	Emerging	Verified
S1-4	Claim	P0	Emerging	Verified
S1-5	Claim	P1	Emerging	Verified
S2-1	Claim	P0	Established	Verified
S2-2	Claim	P0	Established	Verified
S2-3	Claim	P1	Established	Verified
S2-4	Claim	P1	Supported	Verified
S3-1	Claim	P1	Supported	Verified
S3-2	Claim	P1	Supported	Verified
S3-3	Claim	P1	Emerging	Verified
S3-4	Argument	P0	Emerging	Verified
S4-1	Claim	P0	Emerging	Verified
S4-2	Claim	P1	Emerging	Verified
S4-3	Claim	P1	Supported	Verified
S4-4	Argument	P1	Emerging	Verified
S5-1	Proposition	P0	Emerging	Verified
S5-2	Claim	P2	Speculative	Verified

source. Verification against the EQUATOR website (accessed 3 March 2026) returned 699 guidelines, prompting a correction to “nearly 700.” The original figure would have understated the landscape by approximately 30%, illustrating how even a straightforward factual claim benefits from structured source checking.

Argument (S3-4). “Claims, arguments, and propositions require fundamentally different verification procedures.” This is the paper’s central argument. Table 4 shows the Toulmin decomposition as applied during verification.

Proposition (S5-1). “The scholarly community should develop verification infrastructure for non-empirical papers.” This is the paper’s single proposition. Its Whetten decomposition: *What*—typed verification infrastructure distinguishing claims, arguments, and propositions. *How*—per-type checklists adapted from Toulmin (arguments) and Whetten (propositions), with confidence-to-language mapping. *Why*—EQUATOR provides nearly 700 guidelines for empirical research (S2-1) while non-empirical papers have none (S2-2), and AI writing amplifies the verification challenge. *Boundary conditions*—applies to academic papers intended for peer-reviewed publication; excludes informal writing, journalism, and creative writing; the proposed checklists are grounded in Western argumentation

Table 4: Toulmin decomposition of the central argument (S3-4).

Component	Content
Claim	Different unit types require different verification procedures.
Grounds	A registry entry classified as a source-backed claim can be scored at low confidence for lacking a citation, even when its underlying reasoning would satisfy a Toulmin-style argument checklist (premises verified, warrant logically valid).
Warrant	If applying the wrong verification procedure to a correctly reasoned entry produces a false failure, then type-specific procedures are necessary to avoid systematic misclassification.
Qualifier	Emerging—argued on design rationale; not independently evaluated across multiple paper projects.
Rebuttal	The argument rests on a structural illustration; a larger sample across domains would be needed to establish generalisability.

traditions and may need adaptation for other scholarly conventions.

Corrections discovered during verification

The verification process caught two categories of issue before submission: (1) the EQUATOR guideline count was corrected from approximately 500 to nearly 700 (S2-1); and (2) the citation hallucination claim (S1-1) initially lacked a source, prompting a literature search that identified Mugaanyi et al. (7). These corrections—ranging from a factual error to missing evidence—are the kinds of issues the framework is designed to catch.

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